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Unit – 4 : Session – 12 : SLO – 2 | Influence on Instruction Sets — Fill in the Blanks

■ I. Fill in the Blanks

1. RISC — a small set of simple, fixed-length instructions.
2. CISC — CISC instruction sets allow memory-to-memory operations for compact code.
3. "make the common case fast" (Amdahl's) — frequently used instructions should be fast.
4. SIMD (vector) — modern processors use SIMD extensions for parallel data processing.
5. code density — embedded systems prioritize compact code to minimize memory usage.
6. fixed-length — the fixed-length format simplifies pipelining but may reduce code density.
7. Predicated — predicated execution conditionally executes instructions, removing branches.
8. Backward — backward compatibility forces x86 to keep complex legacy features.
9. load-store — the RISC load-store model requires explicit load/store instructions for memory access.
10. Multiple (complex) — many addressing modes increase flexibility but complicate hardware design.